

R Essentials

Data Import & Summary

```
data <- read.csv("file.csv")  
summary(data)  
head(data)  
nrow(data)
```

Basic Statistics

```
mean(data$variable)  
median(data$variable)  
sd(data$variable)  
var(data$variable)  
quantile(data$variable, c(0.25, 0.75))
```

Plotting

```
boxplot(variable ~ group, data=data)  
hist(variable)  
plot(x, y)
```

Data Manipulation

Converts wide to long format

```
lengths <- c(data$X, data$Y)  
species <- c(rep("X", length(data$X)),  
             rep("Y", length(data$Y)))  
new <- data.frame(length=lengths,  
                  species=species)
```

Statistical Tests

t-test (2 groups)

```
t.test(variable ~ group, data=data)  
 $H_0: \mu_1 = \mu_2$      $H_1: \mu_1 \neq \mu_2$ 
```

ANOVA (3+ groups)

```
aov(variable ~ group, data=data)  
 $H_0$ : All means equal  
 $H_1$ : At least one mean differs
```

Linear Regression

```
model <- lm(y ~ x, data=data)  
summary(model)  
 $H_0: \beta = 0$  (no relationship)  
 $H_1: \beta \neq 0$  (relationship exists)
```

Chi-squared

```
chisq.test(table)  
chisq.test(observed, p=expected_prob)  
 $H_0: \beta = 0$  (no associations)  
 $H_1: \beta \neq 0$  (has associations)
```

Correlation

```
cor.test(x, y)  
 $H_0: \beta = 0$  (no correlation)  
 $H_1: \beta \neq 0$  (has correlation)
```

Bootstrap (for confidence interval)

```
boot <- replicate(1000, {  
  s = sample(data$variable,  
             size=length(data$variable),  
             replace=TRUE)  
  mean(s)  
})
```

```
quantile(boot, c(0.025, 0.975))
```

Permutation (for hypothesis testing)

Purpose: Test if groups differ (P-values)
Method: Shuffle group labels

Observed difference

```
observed_diff <- mean(data$Y) -  
  mean(data$X)
```

Permutation test

```
sim <- replicate(1000, {  
  shuffled <- sample(data$species)  
  mean(data$length[shuffled=="Y"])  
  - mean(data$length[shuffled=="X"])  
})
```

```
mean(abs(sim)) >= abs(observed_diff)
```

P-values

$P\text{-value} \leftarrow \text{mean}(\text{abs}(\text{sim}) \geq \text{abs}(\text{obs_diff}))$

Interpretation Guide

P-value Interpretation

- $P < 0.05$: "Strong evidence against H_0 "
- $P \geq 0.05$: "No evidence against H_0 "

Box Plot Comparisons

1. Central tendency: Compare medians
2. Variability: Compare IQR (box width)
3. Outliers: Note points beyond whiskers
4. Distribution Shape: Symmetric vs skewed

Example: "Species Y has a higher median length than X. Species X shows greater variability with wider IQR."

Bootstrap vs Permutation

- Bootstrap: Estimates uncertainty (confidence intervals)
 - Resamples WITH replacement
 - Preserves original relationships
- Permutation: Tests hypotheses (P-value)
 - Shuffles group labels
 - Breaks relationships to simulate H_0

Test Selection Guide

- 2 groups, continuous: t-test
- 3+ groups, continuous: ANOVA
- Categories: Chi-square
- Relationship: Correlation / Regression
- Multiple quizzes: ANOVA + post-hoc

Distribution Formulas

Binomial: n trials, probability p

- Mean = np , Var = $np(1-p)$
- Use when: fixed trials, 2 outcomes

Poisson: Rate λ events per time

- Mean = Var = λ
- Use when: rare events, rare constant

Normal:

- 68% within 1 SD, 95% within 2 SD
- CLT: $\bar{x} \sim N(\mu, \sigma^2/n)$

Confidence Interval

- 95% CI: 2.5th to 97.5th Percentile
- Doesn't include 0: Significant diff
- Includes 0: No evidence of diff

Formulas

- $SE = SD / \sqrt{n}$
- $t = (\bar{x} - \mu) / (SE)$
- $\chi^2 = \sum (O - E)^2 / E$
- r = correlation coefficient (-1 to 1)
- R^2 = proportion of variance explained